

SQL PROJECT • DATA ANALYSIS

Air Cargo Analysis

Exploring airline operations, ticketing and revenue through structured SQL queries

4

RELATIONAL TABLES

18

SQL TASKS SOLVED

\$15,369

REVENUE ANALYZED

What This Project Covers

A MySQL database (AirCargo) built from four linked datasets — customers, ticket sales, passenger flight records, and routes — used to answer 18 real business questions with SQL.

1

Database Design

Created the AirCargo database and an ER diagram linking all four tables.

2

Core Queries

Filtering, joins, subqueries and string functions to profile customers and travel.

3

Advanced SQL

Window functions, ROLLUP, indexing, views and a stored procedure.

4

Administration

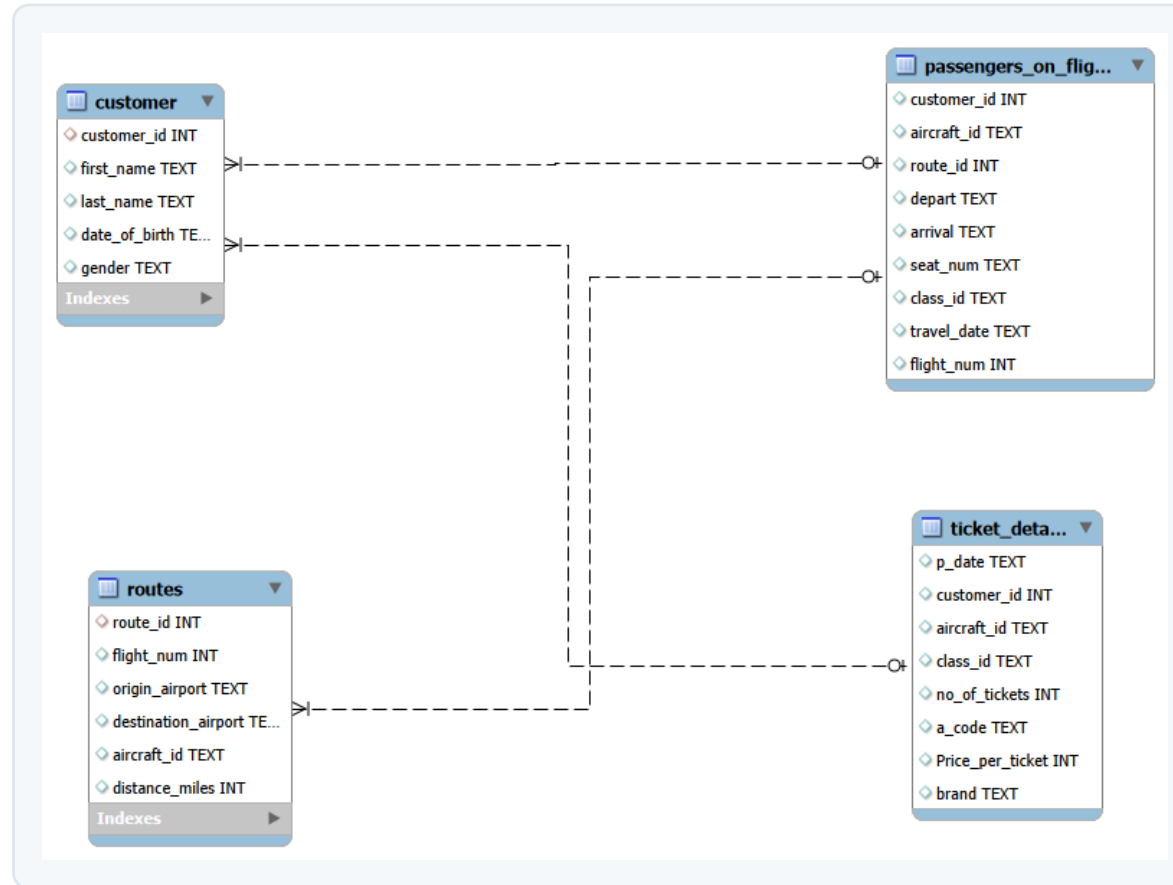
User creation, privilege grants, and execution-plan analysis.

DATASETS USED

- **customer.csv**
50 customer records
- **ticket_details.csv**
50 bookings, 4 travel classes
- **passengers_on_flights.csv**
50 flight-passenger records
- **routes**
Origin, destination & distance

Entity-Relationship Diagram

Four tables connected through customer_id, route_id and aircraft_id keys.



The Numbers at a Glance

50

CUSTOMERS REGISTERED

50

TICKETS BOOKED

\$15,369

TOTAL TICKET REVENUE

4

TRAVEL CLASSES

5

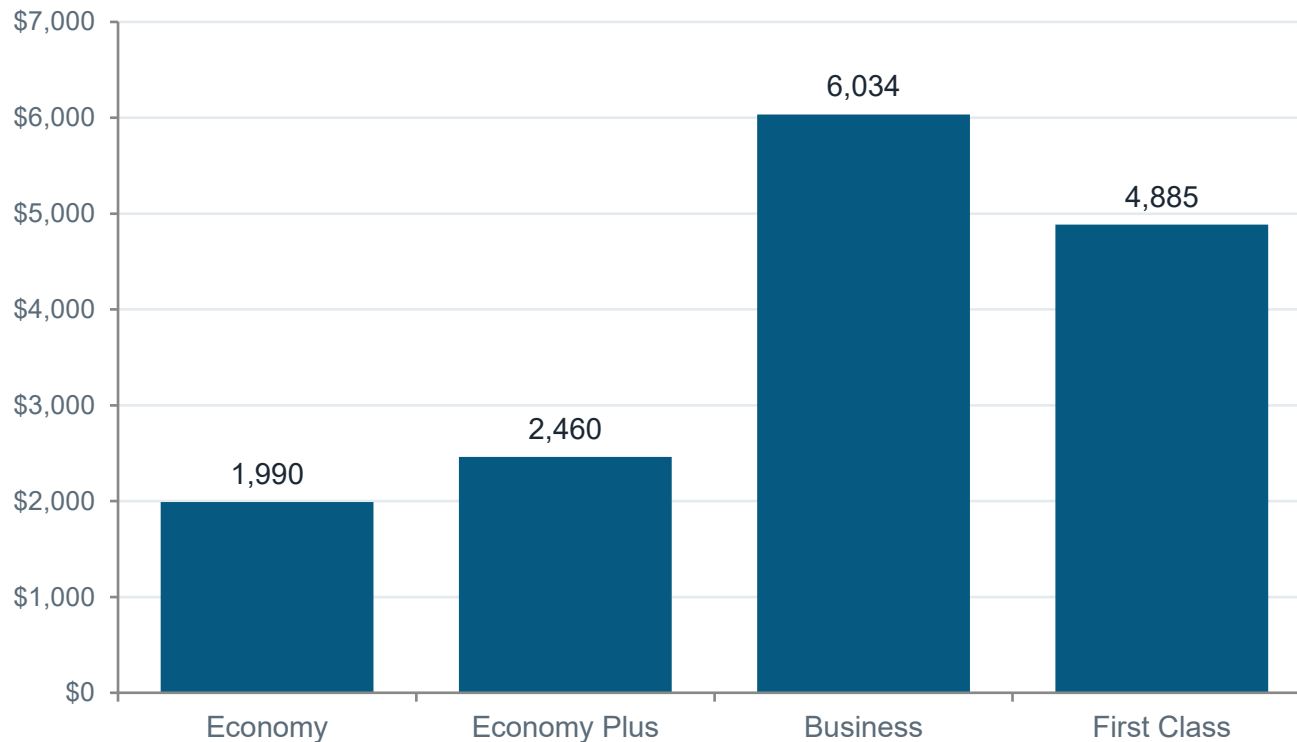
AIRLINE BRANDS

32

FLIGHT ROUTES

Revenue by Travel Class

Business and First Class together generate over 70% of total ticket revenue, despite carrying under half the passengers.



SQL TASK #4

Business Class passengers & revenue

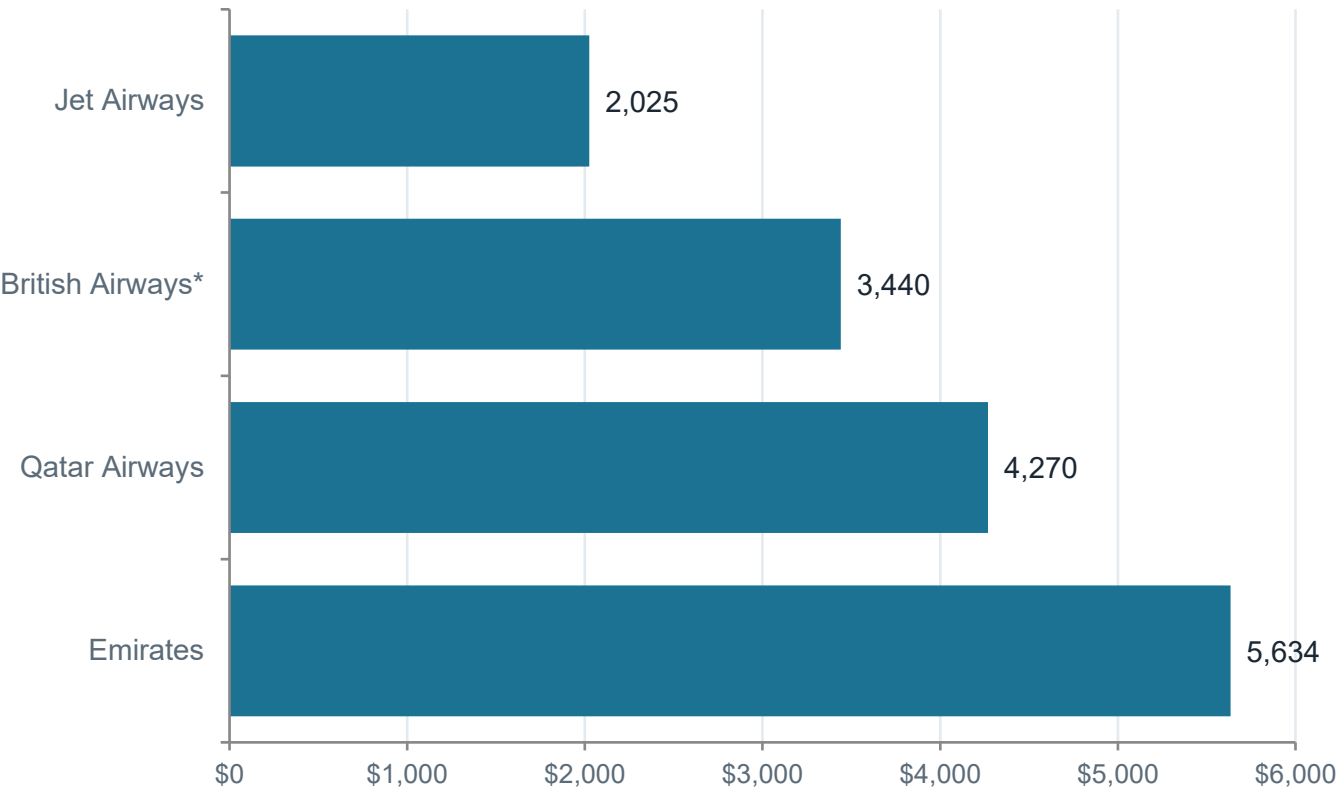
```
SELECT
    SUM(no_of_tickets) AS total_passengers,
    SUM(no_of_tickets *
        Price_per_ticket) AS total_revenue
FROM ticket_details
WHERE class_id = 'Business';
```

Result

13 passengers • \$6,034 revenue

Revenue by Airline Brand

Emirates leads all carriers in booked revenue, followed closely by Qatar Airways.



*Combines two spelling variants of British Airways found in the source data.

SQL TASK #7

Customers who flew Emirates

```
SELECT c.customer_id,  
       c.first_name, c.last_name,  
       t.brand  
FROM customer c  
JOIN ticket_details t  
  ON c.customer_id = t.customer_id  
WHERE t.brand = 'Emirates';
```

A customer–ticket join filtered to one brand — the same join pattern used to link registered customers to their bookings.

Filtering, Joins & Subqueries

01

Range Filtering

Passengers traveling on routes 1 to 25, using BETWEEN on route_id.

02

String Functions

CONCAT(first_name, last_name) builds each customer's full name.

03

Inner Join

customer ⋈ ticket_details links registrations to actual bookings.

04

Subquery

Customers who flew Economy Plus, found via a subquery on passengers_on_flights.

05

Conditional Logic

IF() flags whether total revenue has crossed the \$10,000 mark.

06

Aggregation

GROUP BY customer_id totals tickets bought and amount paid per customer.

Window Functions & Rollups

Two techniques for comparing rows without collapsing the detail.

MAX() OVER (PARTITION BY ...)

Highest ticket price within each class

```
SELECT class_id, price_per_ticket,  
       MAX(Price_per_ticket)  
       OVER (PARTITION BY class_id)  
       AS max_ticket_price  
FROM ticket_details;
```

Every row keeps its detail, while also showing the ceiling price for its own class — no GROUP BY collapse needed.

GROUP BY ... WITH ROLLUP

Ticket totals per customer & aircraft, plus subtotals

```
SELECT customer_id, aircraft_id,  
       SUM(no_of_tickets *  
           Price_per_ticket) AS total_price  
FROM ticket_details  
GROUP BY customer_id, aircraft_id  
WITH ROLLUP;
```

ROLLUP adds automatic subtotal rows per customer and a grand total row — one query instead of three.

Indexes, Views & Stored Procedures



Index

```
idx_route_id
```

Speeds up lookups filtered by route_id, e.g. finding all passengers on route 4.



Execution Plan

```
EXPLAIN
```

Confirms the index is used instead of a full table scan for route 4.



View

```
business_class_customers
```

A saved query exposing only customer_id, class_id and brand for Business class.



Stored Procedure

```
Get_Long_Distance_Routes()
```

Returns every route over 2,000 miles — callable on demand, no query rewriting.

User Access & Security

A dedicated database user was created and scoped to the AirCargo schema only.

```
CREATE USER 'aircargo_user'@'localhost'  
  IDENTIFIED BY 'AirCargo@123';  
  
GRANT ALL PRIVILEGES  
  ON AirCargo.*  
  TO 'aircargo_user'@'localhost';  
  
FLUSH PRIVILEGES;
```

WHY IT MATTERS

- Least-privilege access instead of shared root logins
- Privileges scoped to a single schema (AirCargo.*)
- FLUSH PRIVILEGES applies grants immediately

What the Data Shows

1 Business & First Class drive most revenue — \$10,919 of \$15,369 total (71%) from just 26 of 50 tickets.

2 Emirates is the top-earning brand at \$5,634, ahead of Qatar Airways at \$4,270.

3 Every registered customer in the sample has also booked a ticket — a fully matched customer base.

4 Route-level demand is thin (avg. ~1.6 passengers/route), suggesting scope to consolidate or promote low-traffic routes.

Thank You

Air Cargo Analysis — SQL Project

Database design • 18 SQL queries • Views, procedures & performance tuning